

MOVIE SALES ANALYSIS & PERFORMANCE REPORT

Box-Office Performance Analysis, 2006-2016

Data Analytics Department | Built in Power BI (DAX, interactive filtering, drill-down)

1. PROJECT OVERVIEW & BUSINESS PROBLEM STATEMENT

This report analyzes ten years of IMDb box-office data (2006-2016) to understand what actually drives movie revenue - release timing, genre, director track record, and the interplay between audience rating, vote volume, and runtime. The goal is straightforward: separate what looks good on paper (ratings, critical scores) from what actually moves box-office numbers, and turn that into a handful of decisions worth acting on.

2. EXECUTIVE KPI PERFORMANCE SUMMARY

Metric / KPI	Value	What it tells us
Total Movies Analyzed	999	The full catalogue behind every figure in this report.
Total Box Office Revenue	\$72,337.96M	Combined revenue across all titles - the baseline every genre and director comparison is measured against.
Average Revenue per Movie	\$82.96M	The typical title earns far less than this - revenue is concentrated in a small number of hits (see Q7).
Average IMDb Rating	6.72 / 10.00	Healthy overall, but slipping year over year (Q2).
Average Runtime	113.17 min	Longer films tend to earn more and rate higher (Q8) - worth keeping in mind at the greenlight stage.
Average Critic Metascore	58.99 / 100	Middling on average; a few genres consistently beat it by a wide margin (Q5).

3. EMPIRICAL BUSINESS QUESTIONS & ROOT-CAUSE ANSWERS (PART I)

Q1: How have releases and revenue trended over the years?

Source Visual: Movies Released & Total Revenue by Year

Both climbed sharply toward the end of the decade. 2016 alone produced 297 titles - more than double 2015's 127 - and brought in \$11.2B, the highest of any year, followed by 2015 (\$8.85B) and 2014 (\$8.0B). The industry (or at least this dataset) was clearly scaling up output year after year.

Q2: What happened to average rating and vote counts?

Source Visual: Avg Rating & Avg Votes by Year

Both drifted down as volume went up. Average rating slipped from 7.13 in 2007 to 6.44 in 2016, and average votes per film fell even harder - from about 269K in 2006 to just 48.6K in 2016. Part of that is simply newer movies not having had time to accumulate votes yet, but the rating trend is a real pattern worth watching.

Q3: Which year led on revenue, and which years released the most movies?

Source Visual: Top Revenue Year & Top 5 Years by Release Count

2016 wins both categories - highest revenue and highest release count. Behind it: 2015, 2014, 2013, and 2012, each releasing fewer movies than the year that followed. Output basically grew every year through this stretch.

Q4: Which genres bring in the most money, and which get the best ratings and engagement?

Source Visual: Revenue by Genre (Treemap); Avg Rating & Total Votes by Genre

Adventure (\$38.9B) and Action (\$35.6B) are the clear revenue leaders, well ahead of Drama and Comedy. But they're not the best-loved genres - War, Animation, and Biography rate highest with audiences (7.2-7.4), despite modest box office. Drama, Action, and Adventure do lead on total votes, so the biggest earners are still getting plenty of attention - just not the top ratings.

3. EMPIRICAL BUSINESS QUESTIONS & ROOT-CAUSE ANSWERS (PART II)

Q5: Which genres earn the strongest reviews from critics?

Source Visual: Avg Metascore by Genre

History (70.9) and Animation (70.8) top the list, with Biography (68.9) and War (64.5) not far behind - all well above the 58.99 dataset average. Same story as Q4: the genres critics love most aren't the ones filling theaters.

Q6: Who are the standout directors and titles?

Source Visual: Top 10 Directors by Revenue; Top-Voted & Top-Grossing Movies

J.J. Abrams, David Yates, and Christopher Nolan lead all directors by career revenue in this dataset (\$1.68B, \$1.63B, and \$1.52B). *The Dark Knight* and *Inception* pull in the most votes (1.79M and 1.58M) and also rate highly (9.0 and 8.8). The single highest earner, *Star Wars: The Force Awakens* (\$936.6M), rates lower at 8.1 - a reminder that box-office size and audience love don't always move together.

Q7: How much of total revenue comes from just the top movies?

Source Visual: Revenue Progression: All Movies to Top 10

A lot. The ten highest-grossing titles bring in \$5.86B - about 8% of all revenue in the dataset, from 1% of the titles. A handful of hits are doing a disproportionate share of the work.

Q8: What actually predicts revenue - rating, votes, or runtime?

Source Visual: Revenue vs Rating vs Votes; Avg Revenue & Rating by Runtime

Votes, by a wide margin. The correlation between votes and revenue (0.64) is nearly three times stronger than rating's (0.22) - buzz and reach matter more than perceived quality. Runtime matters too: movies over 140 minutes average \$161M and a 7.5 rating, more than triple the revenue of anything under 90 minutes (\$45M, 5.9 rating). Longer, bigger productions simply tend to perform better on both fronts.

4. KEY FINDINGS & BUSINESS INSIGHTS

- **Output grew fast, and quality metrics slipped a little as it did.** More movies, more revenue, but slightly lower average ratings each year - worth watching, not necessarily a red flag on its own.
- **Critical love and box office don't overlap much.** The best-reviewed genres (War, Animation, Biography, History) aren't the ones making the most money (Adventure, Action) - the two barely correlate.
- **Buzz beats quality as a revenue signal.** Vote count predicts revenue far better than rating does - reach and word of mouth matter more than being "good."
- **A small number of films carry a lot of the revenue.** The top 10 titles bring in 8% of everything - performance here rests on a handful of hits, not the catalogue as a whole.

5. STRATEGIC RECOMMENDATIONS

- **Weigh reach alongside quality when picking projects.** Vote volume tracks revenue better than rating does, so audience-reach potential deserves real weight in greenlighting decisions.
- **Lean into longer runtimes where the story supports it.** The 140+ minute tier outperforms on both revenue and rating - a reasonably safe bet for bigger releases.
- **Keep a foot in both camps on genre.** Action/Adventure for revenue, but don't starve Animation/Biography/History - they protect the brand's critical reputation even when the box office is smaller.
- **Plan for hit-driven revenue, not steady revenue.** With 8% of revenue riding on 10 titles, one underperforming tentpole can move the whole year's numbers.
- **Study what made 2016 work.** It's the standout year on both volume and revenue - worth digging into what drove that before assuming it's just a data-recency artifact.

How this was built: The Power BI report runs on a single IMDb movie table (title, genre, director, actors, year, runtime, rating, votes, revenue, metacore), with genre and actor fields split out for category-level breakdowns. It spans four pages - Movie Overview & Time Trends, Genre & Rating Analysis, Top Performers & Box Office Insights, and Correlation & Advanced Metrics - using cards, column/line/bar charts, a treemap, scatter plot, waterfall chart, matrix, and slicers for interactivity.